

MPG No. 8.3.26 $m'mm$ (m'mm setting) [Type III, orthorhombic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & 0 \end{bmatrix}$$

$$T_x = T_{px}^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 2T_{fax}^{(1)} + 3T_{px}^{(1)}$$

$$T_{yyx} = -T_{fax}^{(1)} + T_{fbx}^{(1)} + T_{px}^{(1)}$$

$$T_{zzx} = -T_{fax}^{(1)} - T_{fbx}^{(1)} + T_{px}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{yyx} = 2M_{dyz}^{(1)} + 2T_{px}^{(2)}$$

$$T_{zzx} = -2M_{dyz}^{(1)} + 2T_{px}^{(2)}$$

$$\frac{T_{zzx}}{2} = -M_{dyz}^{(1)} + T_{px}^{(2)}$$

$$\frac{T_{yyx}}{2} = M_{dyz}^{(1)} + T_{px}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{zzx} = M_{dyz}^{(2)} + T_{px}^{(3)}$$

$$T_{xyy} = M_{dyz}^{(2)} - T_{px}^{(3)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).